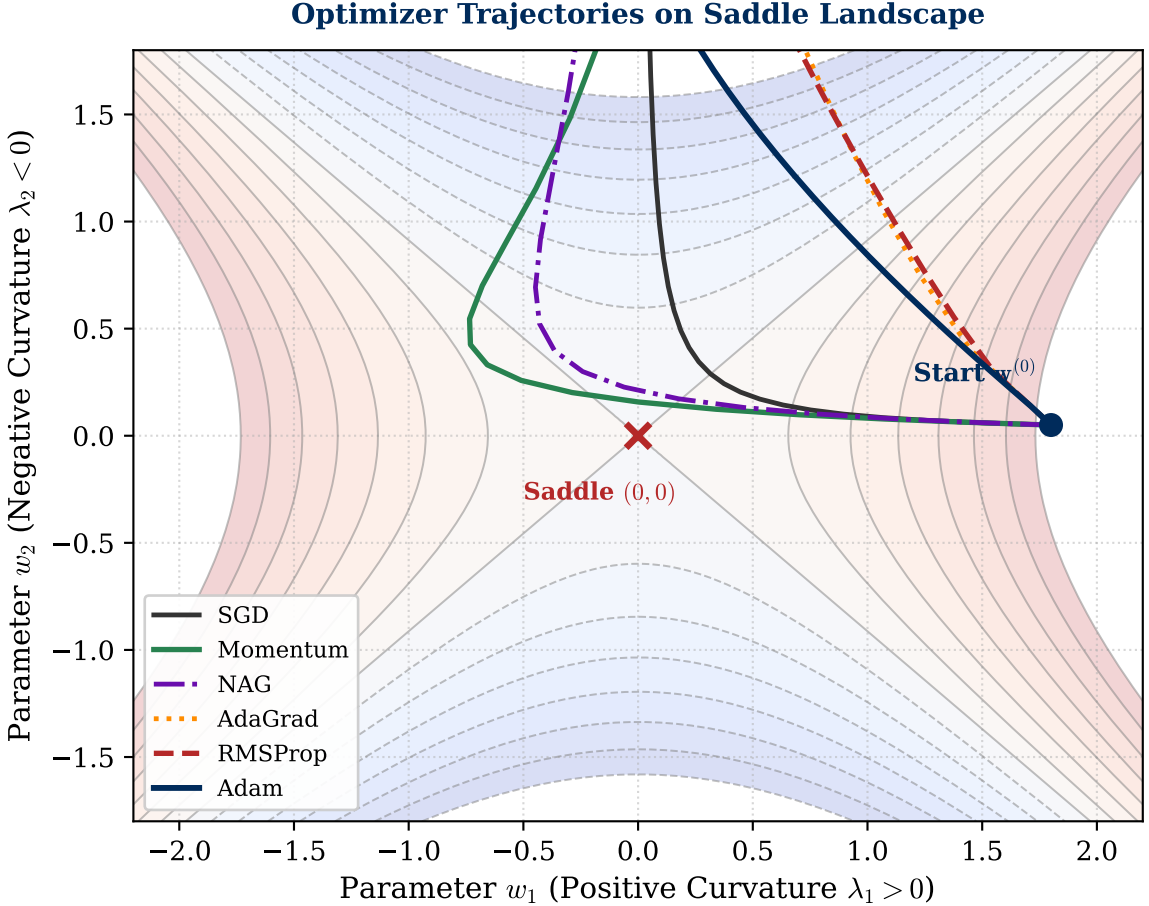

Optimizer Trajectories on Saddle Landscape
Start $\mathbf{w}^{(0)}$
Saddle $(0, 0)$
SGD
Momentum
NAG
AdaGrad
RMSProp
Adam
Parameter $w_2$ (Negative Curvature $\lambda_2 < 0$)
Parameter $w_1$ (Positive Curvature $\lambda_1 > 0$)
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0